ARTIN STATUS ACROSS CONSECUTIVE PRIMES AND DISTINCT BASES: MIXED EXCLUSION LAWS AND THE COMPLETION OF A 2×2 PROGRAMME

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Abstract. Say a prime p is Artin base a if a is a primitive root modulo p. This paper completes a two-axis programme on the joint statistics of Artin status: having previously treated one base at consecutive primes, and several bases at one prime, we now measure the *mixed* correlations $\delta(a \to b) = P\big(p_{n+1} \in \mathrm{A}_b \mid p_n \in \mathrm{A}_a\big) - P\big(p_{n+1} \in \mathrm{A}_b \mid p_n \notin \mathrm{A}_a\big)$ for all 144 ordered pairs from twelve bases, over all 50,847,531 consecutive prime pairs below 10^9. On the deterministic side we classify the *mixed exclusion classes*: gap classes g modulo $\mathrm{lcm}(f_a, f_b)$ at which no admissible residue permits $\chi_a(p) = \chi_b(p+g) = -1$, so that p_n Artin base a and p_{n+1} Artin base b can never co-occur. Genuinely mixed instances exist — for the ordered pairs $(2,6)$ and $(6,2)$ the classes $g \equiv 10, 14 \pmod{24}$ are excluded although neither base excludes them alone — and the classification is verified on 104,972,810 pairs with zero exceptions and confirmed complete by a data-driven converse scan. On the statistical side, the off-diagonal correlations are small but overwhelmingly significant (124 of 132 with $|z| > 3$), negative for 101 of 132 ordered pairs (mean -0.0035), an order of magnitude weaker than the diagonal ones — and *asymmetric*: $\delta(a \to b) \neq \delta(b \to a)$, with differences up to 0.05, reflecting the asymmetric conditioning through two distinct integers $p_n - 1$ and $p_{n+1} - 1$. The strongest organising variable for $|\delta(a \to b)|$ is the entanglement of the base pair, followed by the conductor of the *second* base. As a byproduct of extending the base set we exhibit base 15 as the first known base whose consecutive-prime Artin correlation is globally *positive* ($\delta = +0.0122$, $z = 87$): its three exclusion classes are outweighed by strongly coupled classes, resolving in the negative the natural conjecture that $\delta(a) < 0$ for every base.

1. Introduction

Let A_a denote the set of primes for which the integer a is a primitive root. Artin's conjecture describes the density of A_a; Hooley proved it under GRH [4]. This is the fourth paper in a series studying the *joint* statistics of Artin status computationally at the scale $x = 10^9$, and it completes a 2×2 programme:

	same base	different bases
same prime	(trivial)	paper III [3]
consecutive primes	papers I, II [1, 2]	**this paper**

Papers I and II found that Artin status of a fixed base a at consecutive primes is anticorrelated, identified deterministic *exclusion laws* — gap classes at which two consecutive primes can never both be Artin base a — and classified them via two mechanisms: reversal of the quadratic character χ_a, and inadmissibility of the residues that would be needed for a doubly-negative configuration. Paper III found that Artin statuses of *different* bases at the *same* prime attract

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(the shared factorisation of $p-1$ is a common cause), except on multiplicatively dependent pairs, where quadratic entanglement produces repulsion and, inside multiplicative triples, a deterministic three-way exclusion.

The remaining axis — base a at p_n against base b at p_{n+1} — mixes the two settings: the conditioning passes through two *distinct* integers p_n-1 and $p_{n+1}-1$ (as in papers I–II), but the two events involve *distinct* quadratic characters χ_a, χ_b (as in paper III). Both the deterministic and the statistical questions have new content here.

Deterministic part. Call a residue class g modulo $L = \mathrm{lcm}(f_a, f_b)$ (where f_a is the conductor of χ_a) a *mixed exclusion class* for the ordered pair (a,b) if there is no residue r with r and $r+g$ admissible (coprime to L) and $\chi_a(r) = \chi_b(r+g) = -1$. Since the Artin property forces the character value -1, primes $p_n \equiv r$, $p_{n+1} = p_n + g'$ with $g' \equiv g \pmod{L}$ can then never satisfy both "a is a primitive root of p_n" and "b is a primitive root of p_{n+1}". Diagonal instances ($a=b$) are the exclusion laws of papers I–II. The classification here (Section 2) yields genuinely mixed instances; the simplest is:

Theorem 1 (see Section 2)**.** *For the ordered pairs* $(2,6)$ *and* $(6,2)$*, the gap classes* $g \equiv 10, 14 \pmod{24}$ *are mixed exclusion classes: no consecutive primes* $p_n, p_{n+1} = p_n + g$ *with* $g \equiv 10, 14 \pmod{24}$ *have* 2 *a primitive root of one and* 6 *a primitive root of the other. Neither base excludes these classes by itself.*

At $x = 10^9$ the predicted mixed exclusion classes across all 144 ordered pairs contain 104,972,810 consecutive prime pairs and exactly zero co-occurrences, and a data-driven converse scan (every class of every pair with at least 5×10^5 pairs and a zero count) finds nothing the classification misses.

Statistical part. Define the mixed correlation

$$\delta(a \to b) \;=\; P\big(p_{n+1} \in \mathrm{A}_b \mid p_n \in \mathrm{A}_a\big) - P\big(p_{n+1} \in \mathrm{A}_b \mid p_n \notin \mathrm{A}_a\big).$$

The diagonal $\delta(a \to a)$ reproduces papers I–II exactly (a useful end-to-end validation, since the present computation is an independent implementation). Off the diagonal we find (Section 4):

(1) *Sign and size.* 101 of the 132 ordered off-diagonal pairs have $\delta(a \to b) < 0$, with mean -0.0035 — repulsion persists across bases, but at an order of magnitude weaker than on the diagonal (mean -0.034 over the eleven bases with negative diagonal). The residue channels that transmit the Lemke Oliver–Soundararajan bias [5] are mostly *different* for χ_a and χ_b, so most of the effect cancels.

(2) *Significance.* 124 of 132 off-diagonal correlations have $|z| > 3$; the structure is small but real.

(3) *Asymmetry.* $\delta(a \to b) \neq \delta(b \to a)$: the mean absolute difference is 0.009 and the maximum 0.051 (attained by $(6,10)$ versus $(10,6)$). No same-prime analogue of this exists — Paper III's $\phi(a,b)$ is symmetric by construction — and the asymmetry witnesses the ordered conditioning through $p_n - 1$ versus $p_{n+1} - 1$.

(4) *Extremes are entangled or share structure.* The most negative mixed correlations are $\delta(6 \to 2) = -0.0354$ and $\delta(21 \to 5) = -0.0330$; the most positive are $\delta(6 \to 10) = +0.0415$ and $\delta(3 \to 15) = +0.0314$. All four involve multiplicatively dependent pairs. Across all 132 ordered pairs, entanglement of $\{a,b\}$ correlates with $|\delta(a \to b)|$ at $r = 0.36$, the strongest single organising variable we find, followed by the conductor of the *second* base ($r(\log f_b, |\delta|) = -0.29$); the conductor of the first base is essentially irrelevant ($r = 0.07$).

A base with positive diagonal correlation. Extending the base set to include $a = 15$ (absent from papers I–II) produced a surprise worth reporting on its own: base 15 has

$$\delta(15 \to 15) = +0.0122 \qquad (z = 87.0),$$

the first known base whose consecutive-prime Artin correlation is globally *positive*. Papers I–II measured $\delta < 0$ for all eleven bases examined and conjectured persistence of the negative sign; base 15 refutes the natural extension of that conjecture to all bases. The mechanism is the class-competition phenomenon identified in paper II: base 15 has three exclusion classes modulo 60 (forcing δ negative on 9.4% of pairs), but its remaining gap classes are dominated by strongly positively coupled ones (e.g. $\delta = +0.64$ at $g \equiv 10 \pmod{60}$, $+0.26$ at $g \equiv 18$), and at conductor 60 — the largest in our set — the negative within-class residue effect is too weak to overcome them. The sign of $\delta(a)$ is thus not universal: it is the resolution of a competition that either side can win, and 15 is the smallest base where the positive side wins.

2. The mixed exclusion classification

Throughout, $d_a = \mathrm{sqf}(a)$, f_a is the conductor of $\chi_a = \left(\frac{d_a}{\cdot}\right)$ (namely d_a if $d_a \equiv 1 \bmod 4$, else $4d_a$), and $L = \mathrm{lcm}(f_a, f_b)$. Admissibility of r means $\gcd(r, L) = 1$; residues attained by primes $p \nmid 2ab$ are exactly the admissible ones.

Definition 2. The class $g \bmod L$ is a *mixed exclusion class* for the ordered pair (a, b) if no admissible r with $r + g$ admissible has $\chi_a(r) = -1$ and $\chi_b(r + g) = -1$.

Proposition 3. *If g is a mixed exclusion class for (a, b), then there are no primes $p, q = p + g'$ with $p \nmid 2ab$, $q \nmid 2ab$, $g' \equiv g \pmod{L}$, such that a is a primitive root modulo p and b is a primitive root modulo q.*

Proof. $\chi_a(p)$ depends only on $p \bmod f_a$, hence on $p \bmod L$; similarly for $\chi_b(q)$. If both primitive-root conditions held, then $\chi_a(p) = \chi_b(q) = -1$ with $p \bmod L$ admissible and $q \equiv p + g$, contradicting Definition 2. □

The classification of mixed exclusion classes reduces, exactly as in paper II, to the behaviour of the prime-discriminant components of χ_a and χ_b under the shift, together with the inadmissibility counting at odd prime conductors; we do not repeat the component analysis, but record the outcome of the exhaustive scan for our twelve bases $\{2, 3, 5, 6, 7, 10, 11, 13, 15, 17, 21, 29\}$: the ordered pairs possessing mixed exclusion classes are the twelve diagonal pairs (with the classes found in papers I–II, including the base-5 inadmissibility classes and both classes of base 21), together with exactly one unordered off-diagonal pair:

Theorem 4. *The ordered pairs $(2, 6)$ and $(6, 2)$ have the mixed exclusion classes $g \equiv 10, 14 \pmod{24}$; no other off-diagonal pair from the twelve bases admits any mixed exclusion class.*

Proof for $(2, 6)$; the converse half is the exhaustive scan. χ_2 has conductor 8 with $\chi_2(r) = -1$ exactly for $r \equiv 3, 5 \pmod 8$; χ_6 has conductor 24 with $\chi_6(r) = -1$ exactly for $r \equiv 7, 11, 13, 17 \pmod{24}$. Fix $g \equiv 10 \pmod{24}$. The residues $r \bmod 24$ with $\chi_2(r) = -1$ are $r \equiv 3, 5, 11, 13, 19, 21$; discarding those not coprime to 24 leaves $r \equiv 5, 11, 13, 19$. Their shifts $r + 10$ are $15, 21, 23, 5 \pmod{24}$; the first two are inadmissible ($3 \mid 15, 21$), and $\chi_6(23) = \chi_6(5) = +1$. So no admissible configuration has both characters -1. The case $g \equiv 14$ is symmetric ($r + 14 \equiv 19, 1, 3, 9$; admissible ones are $19, 1$, and $\chi_6(19) = \chi_6(1) = +1$). Note the essential role of inadmissibility: on the full residue system the configurations would exist, but they land on multiples of 3. □

TABLE 1. Verification of mixed exclusion classes at $x = 10^9$. Ordered pairs (a, b): a tested at p_n, b at p_{n+1}. Diagonal rows reproduce papers I–II (including the inadmissibility classes of base 5); the $(2, 6)/(6, 2)$ rows are the genuinely mixed law of Theorem 4.

(a,b)	classes mod L	L	pairs	co-occurrences
$(2,2)$	4	8	13,404,106	0
$(3,3)$	$4,6,8$	12	27,010,759	0
$(5,5)$	$2,3$	5	20,523,554	0
$(6,6)$	$8,12,16$	24	12,419,039	0
$(7,7)$	14	28	3,567,335	0
$(10,10)$	20	40	2,214,511	0
$(11,11)$	22	44	1,796,191	0
$(15,15)$	$20,30,40$	60	4,782,744	0
$(21,21)$	$7,14$	21	4,046,375	0
$(2,6)$	$10,14$	24	7,604,098	0
$(6,2)$	$10,14$	24	7,604,098	0
total			104,972,810	**0**

Remark 5. The pair $(2, 6)$ is multiplicatively dependent ($\mathrm{sqf}(2 \cdot 6) = 3$, present in the ambient set), and the mechanism visibly mixes the two ingredients of paper II: a character shift (as in reversal) and residue inadmissibility (as at base 5). That the same two mechanisms exhaust the mixed case as well — confirmed by the converse scan below — supports the view of papers II and III that quadratic-character parity and admissibility are the only deterministic constraints at this level, with all further structure being statistical.

Verification at 10^9 and completeness. For every ordered pair and every predicted class we counted, among the 50,847,531 consecutive prime pairs below 10^9, the pairs whose gap lies in the class and, among them, those with a a primitive root of p_n and b of p_{n+1}. Table 1 reports the nonzero rows. The predicted classes contain 104,972,810 pairs in total and the co-occurrence count is zero in every row. Conversely, a data-driven scan of *all* classes of *all* 144 ordered pairs (modulo the appropriate L) for zero counts supported by at least 5×10^5 pairs returns exactly the predicted classes and nothing else: the classification is complete at the resolution of the data.

3. DATA AND COMPUTATION

The computation is one streaming pass of a segmented sieve to 10^9 (50,847,532 primes ≥ 5; 50,847,531 consecutive pairs). For each prime, $p - 1$ is factored once by trial division and the Artin indicator computed for all twelve bases by the standard criterion ($a^{(p-1)/\ell} \not\equiv 1$ for all prime $\ell \mid p - 1$, 128-bit arithmetic); for each ordered base pair (a, b) we accumulate the 2×2 table of $\big(\mathbf{1}[p_n \in \mathrm{A}_a], \mathbf{1}[p_{n+1} \in \mathrm{A}_b]\big)$, globally and per gap $g \leq 300$. All counts are exact. Hardware and validation are as in the companion papers (2-core Xeon E5-2673 v4, 8 GB RAM; single core); the run takes under an hour. Beyond the usual checks (prime counts; marginal densities matching the Hooley values to four decimals, including 0.393642 for base 5 against $C \cdot 20/19 = 0.393638$), the twelve diagonal correlations reproduce the values of papers I–II to all reported digits from an independently written accumulator, and the verification of Table 1 doubles as a check of the classification.

TABLE 2. Extremes of the off-diagonal mixed correlation $\delta(a \to b)$ at $x = 10^9$. "Dep." marks multiplicatively dependent $\{a, b\}$ (the product is a square times a third base of the set).

a	b	$\delta(a \to b)$	z	Dep.
6	2	-0.0354	-252.6	✓
21	5	-0.0330	-233.1	✓
3	7	-0.0233	-165	✓
11	3	-0.0201	-142	
7	3	-0.0193	-137	✓
2	6	$+0.0094$	$+67.2$	✓
15	3	$+0.0222$	$+158.4$	✓
3	15	$+0.0314$	$+224.2$	✓
6	10	$+0.0415$	$+295.8$	✓

4. The mixed correlation matrix

The full 12×12 matrix is available with the data; we summarise its structure.

Observation 6 (Weak, mostly negative, highly significant)**.** Of the 132 ordered off-diagonal pairs, 101 have $\delta(a \to b) < 0$; the mean is -0.0035, one order of magnitude below the mean -0.034 of the eleven negative diagonal values; and 124 of 132 have $|z| > 3$. As in the companion papers, at this sample size the z-scores certify only that the values are real; the relevant magnitudes are the effect sizes.

The interpretation follows the channel picture of papers I–II: the Lemke Oliver–Soundararajan repulsion acts on the joint residues of (p_n, p_{n+1}), and transmits to the pair $(\mathrm{A}_a(p_n), \mathrm{A}_b(p_{n+1}))$ only through residue classes that both characters (and both ω-channels) see. For $a = b$ every channel is shared; for $a \neq b$ only the common structure — the mutual conductor divisors and the universal $\omega(p-1)$ channel — survives, and the correlation drops by the factor observed.

Observation 7 (Asymmetry)**.** $\delta(a \to b) \neq \delta(b \to a)$ in general: the mean of $|\delta(a \to b) - \delta(b \to a)|$ over unordered pairs is 0.0090 and the maximum is 0.0511, attained by $\{6, 10\}$ ($\delta(6 \to 10) = +0.0415$ versus $\delta(10 \to 6) = -0.0096$).

The asymmetry has no analogue in paper III (where the same-prime $\phi(a, b)$ is symmetric by definition) and directly witnesses the ordering of the conditioning: $\delta(a \to b)$ conditions on the arithmetic of $p_n - 1$ and measures on $p_{n+1} - 1$. Its size — comparable to the off-diagonal effects themselves — shows that the mixed correlation is not a symmetric "base similarity" but a directed transition statistic.

Observation 8 (What organises the magnitudes)**.** Over the 132 ordered pairs: multiplicative dependence of $\{a, b\}$ correlates with $|\delta(a \to b)|$ at $r = +0.36$; the conductor of the second base at $r(\log f_b, |\delta|) = -0.29$ (and $r(\log \operatorname{lcm}(f_a, f_b), |\delta|) = -0.32$); the conductor of the first base is negligible ($r = +0.07$). All four extreme values of Table 2 involve dependent pairs.

That f_b matters and f_a does not is consistent with the conductor law of paper II acting at the *measured* prime: the size of $|\delta(a \to b)|$ is controlled by how concentrated the χ_b-channels are at p_{n+1}, while the conditioning event at p_n enters mainly through whether its channels overlap those of b — which is what multiplicative dependence measures.

5. Base 15: a positive diagonal

Base 15 was not among the eleven bases of papers I–II. Its diagonal correlation at $x = 10^9$ is

$$\delta(15 \to 15) = +0.012196, \qquad z = +87.0,$$

positive — the first base known to us with globally positive consecutive-prime Artin correlation, against the uniform negativity of the other twelve measured so far (this paper adds 15; papers I–II measured eleven others, all negative).

The decomposition by gap class modulo 60 (conductor of χ_{15}) explains the sign. The exclusion classes $g \equiv 20, 30, 40 \pmod{60}$ contain 9.4% of all pairs and force $P(\text{both Artin}) = 0$ there; but among the populous non-excluded classes the coupling is strongly positive: $\delta = +0.64$ at $g \equiv 10$, $+0.26$ at $g \equiv 18$, $+0.24$ at $g \equiv 6, 8$, against -0.20 at $g \equiv 2$ and -0.19 at $g \equiv 12$. In the class-competition language of paper II, base 15 is the first base whose positively coupled classes outweigh the forced zeros and the within-class repulsion. Two structural facts make this possible: the conductor 60 is the largest in the set (so the within-class Lemke Oliver–Soundararajan residue effect, which scales down with conductor, is the weakest), while the character $\chi_{15} = \chi_{-3}\chi_5\chi_{-4}$ has three prime-discriminant components whose joint preservation classes ($g \equiv 0, 10, 50 \pmod{60}$, plus partial preservation elsewhere) are unusually rich. We conjecture that $\delta(a) > 0$ for infinitely many bases — specifically for bases whose conductor is large and whose preserving classes are dense — and note that the sign of $\delta(a)$ is therefore not an intrinsic feature of the Artin property but an arithmetic accident of the base.

6. Discussion

The programme completed. Across the four papers the picture is uniform. On every axis the joint behaviour of Artin status splits into a deterministic part — parity and admissibility constraints on quadratic characters, now classified in all three non-trivial settings (diagonal consecutive, same-prime cross-base, mixed) — and a statistical part carried by the arithmetic of $p-1$: the LOS bias through residue channels for consecutive primes, the shared factorisation for same-prime pairs, and their directed, mostly cancelling combination in the mixed case. The deterministic laws are absolute but concentrated on sparse gap classes; the statistical effects are small but global; and which of them dominates a headline number like the sign of $\delta(a)$ is a competition, as base 15 now demonstrates.

Future work. (1) The channel-by-channel derivation of the conductor law (paper II) extends naturally to the mixed setting and would predict the f_b-dominance of Observation 8; the mixed data provide 132 constraints for such a derivation rather than eleven. (2) A systematic search for further positive-δ bases (candidates: large conductor, many prime-discriminant components — e.g. $a = 30, 35, 39, 105$) would test the conjecture of Section 5. (3) The decay in x of the mixed correlations, and of $\delta(15)$ in particular, remains unmeasured; the sign of $\delta(15)$ at 10^{12} is an especially clean question, since the LOS-driven within-class repulsion decays like $1/\log x$ while the class-composition effect does not.

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Declaration of interest

The author reports there are no competing interests to declare.

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Data availability

All code and results are publicly available at `https://github.com/jpbald93/consecutive-artin`: the one-pass program (`mixed_axis.c`), the symbolic classification scripts, the full 12×12 correlation matrix and per-gap contingency tables, and scripts reproducing every number in this paper. The computation reproduces in under an hour on a single core.

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